

# Mini-Batch Gradient Descent

## Contents

|                                                   |   |
|---------------------------------------------------|---|
| 1. Overview . . . . .                             | 2 |
| 2. A family that contains the other two . . . . . | 2 |
| 3. How it works . . . . .                         | 3 |
| 4. Comparing the three paths . . . . .            | 5 |
| 5. Choosing the batch size . . . . .              | 6 |
| 6. Mini-batch in scikit-learn . . . . .           | 8 |
| 7. Summary . . . . .                              | 9 |
| 8. Sources . . . . .                              | 9 |
| 9. Key terms . . . . .                            | 9 |

**Where this fits:** Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Gradient descent ([Note 57](#)).
- **Leads to:** Batch size in Keras ([Note 1020](#)); Batch normalisation ([Note 1031](#)); Optimizers in deep learning ([Note 1032](#)).
- **Compare with:** Stochastic gradient descent ([Note 59](#)).

## 1. Overview

**Key point:** Mini-batch gradient descent updates the coefficients after each small group of observations. Mini-batch sits between batch (stable, slow) and stochastic (fast, noisy) gradient descent, and is the version most used in practice.

A **feature** (G-772) is an input variable (one column of the data table), an **observation** (G-1374) is one record (one row), and the **target** (G-1949) is the output we predict.

**Gradient descent** (G-862) finds the best **coefficients** (G-407) by repeated small steps downhill on the loss ([Note 57](#)). The two types so far sit at opposite ends. Batch gradient descent reads every observation before each step: steady, but slow. Stochastic gradient descent steps after every single observation: fast, but noisy. The in-between option reads a small group of observations per step.

**Mini-batch gradient descent** (G-1222) is this third type. The three types differ only in how often they update the coefficients in one **epoch** (G-696), one full pass over the training data:

- **Batch:** once per epoch, after looking at all  $n$  observations.
- **Stochastic:**  $n$  times per epoch, after each single observation.
- **Mini-batch:** once after each **batch** (G-263): a small group of observations, typically 32 to 256, sometimes 16. Most deep-learning training works this way (Goodfellow §8.1.3).

With 1,000 observations and a batch size of 100, the data splits into 10 batches, so there are 10 updates per epoch. A batch size of 10 would give 100 updates per epoch.

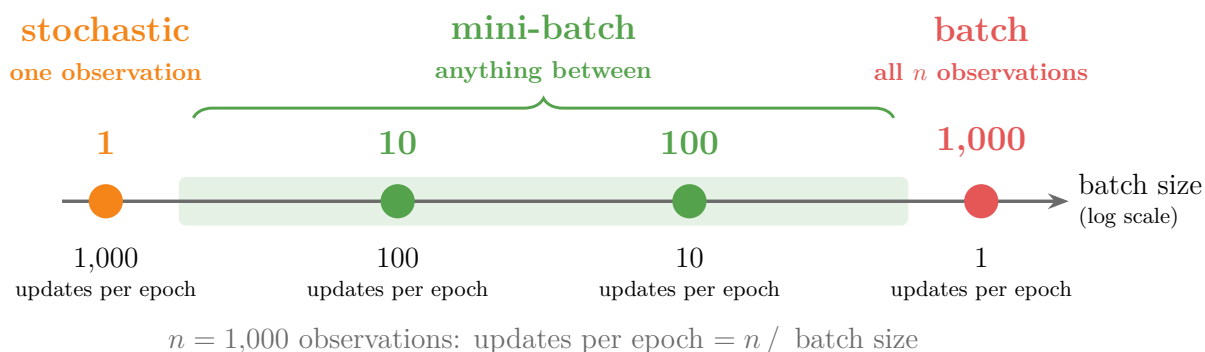
## 2. A family that contains the other two

**Key point:** Batch size  $n$  is batch gradient descent; batch size 1 is stochastic gradient descent; anything between is mini-batch.

The **batch size** (G-267) is a **hyperparameter** (G-910): a setting we choose before training, not one the model learns. Read the table of the three types in the [batch gradient descent Note](#) (section one) by batch size:

- **batch gradient descent** (G-264) is batch size  $n$ ;
- **stochastic gradient descent** (G-1892) is batch size 1;
- **mini-batch** is everything between.

Mini-batch is therefore the general form, and the batch size tunes how it behaves between the two extremes. Figure 1 puts the three names on one batch-size axis for the 1,000 observations of section 1. Watch the updates per epoch fall as the batch size grows.



**Figure 1:** One hyperparameter, three names. With 1,000 observations, batch size 1 is stochastic gradient descent (1,000 updates per epoch), 1,000 is batch gradient descent (1 update), and every size between is mini-batch.

### 3. How it works

**Key point:** Each epoch: shuffle the observations, cut them into batches, and for each batch compute the derivatives from its observations only and update.

In words: instead of asking every observation which way is downhill, or just one, we ask a small group, average their answers, and take one step. With 100 observations and a batch size of 10, an epoch has 10 such steps.

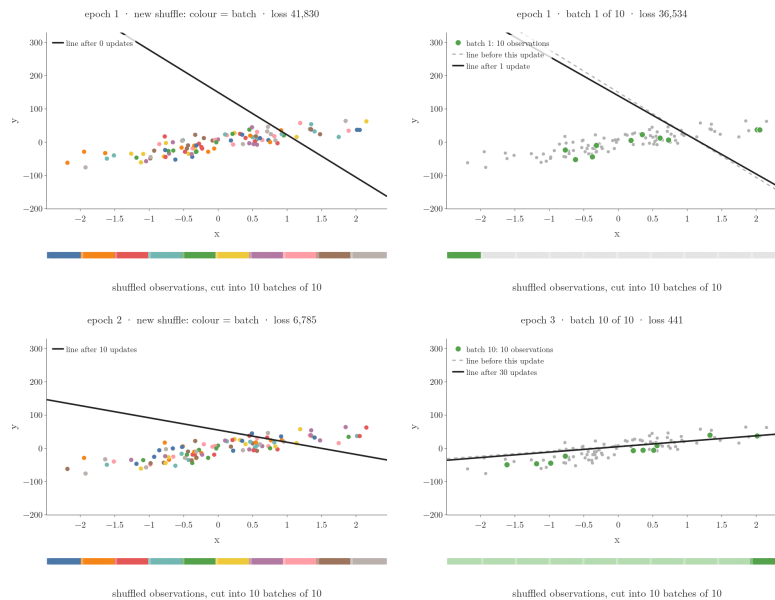
1. Start with any coefficients, for example  $\beta_0 = 0$  and every  $\beta_j = 1$ .
2. For each epoch:
  - put the observations in a new random order, called **shuffling** (G-1797);
  - cut them into consecutive batches of the chosen size (the last batch may be smaller);
  - for each batch, compute the predictions, the errors and the derivatives using only its observations, and update.

The **derivatives** (G-595) of the loss  $L$  are the batch ones, averaged over the observations  $B$  of the current batch instead of all  $n$  observations:

$$\frac{\partial L}{\partial \beta_j} = -\frac{2}{|B|} \sum_{i \in B} (y_i - \hat{y}_i) x_{ij}$$

where  $|B|$  is the number of observations in the batch,  $y_i$  is the target of observation  $i$ ,  $\hat{y}_i$  its prediction and  $x_{ij}$  its value of feature  $j$ . Each coefficient then moves against its derivative, by the derivative times the **learning rate** (G-1068).

Figure 2 runs these steps on the 100-point example of section 4, with batch size 10 and learning rate 0.05, for 3 epochs. Each epoch opens with a new shuffle, which colours every point by the batch it falls into; then one batch at a time lights up green and the line makes one update. Watch the colours change between epochs, and the line move only once per batch.



**Figure 2:** Three epochs of mini-batch gradient descent, batch size 10. Top: the data and the line before (dashed) and after (solid) each update. Bottom: the shuffled observations cut into 10 batches. The loss falls from 41,830 to 6,785 after one epoch and to 441 after three (best possible 283).

**Python:** Mini-batch gradient descent from scratch.

```
class MBGDRegressor:
    def __init__(self, batch_size, learning_rate=0.01,
                  epochs=100, seed=0):
        self.bs, self.lr, self.epochs = (batch_size,
                                           learning_rate, epochs)
        self.rng = np.random.default_rng(seed)

    def fit(self, X, y):
        n = X.shape[0]
        self.intercept_ = 0.0
        self.coef_ = np.ones(X.shape[1])
        for _ in range(self.epochs):
```

```

        order = self.rng.permutation(n)           # shuffle
        for start in range(0, n, self.bs):
            j = order[start:start + self.bs]      # one batch
            err = y[j] - (X[j] @ self.coef_ + self.intercept_)
            self.intercept_ -= self.lr * (-2 * err.mean())
            self.coef_ -= self.lr * (-2 * X[j].T @ err / len(j))
    return self

```

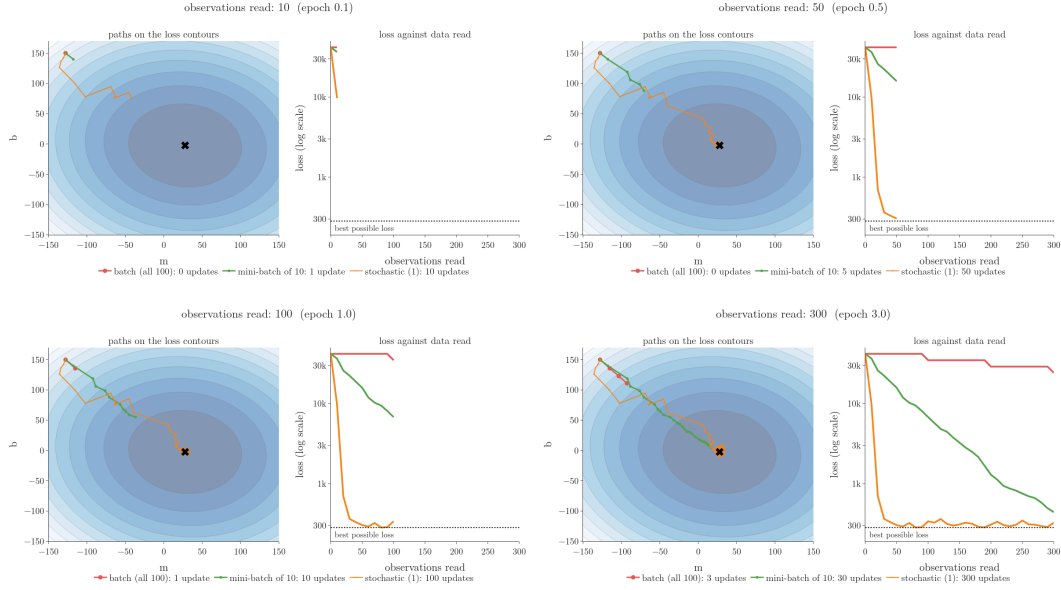
The derivative code is the same vectorised code as batch gradient descent, applied to the observations  $X[j]$  of one batch.

**Extra:** Some implementations draw each batch at random from all observations, so observations can repeat within an epoch. Shuffling and slicing, as above, uses every observation exactly once per epoch. For very large datasets it is usually enough to shuffle the order once and keep it, while never shuffling at all can seriously hurt the result (Goodfellow §8.1.3).

## 4. Comparing the three paths

**Key point:** In three epochs, batch barely moves, stochastic arrives but zigzags, and mini-batch arrives on a smoother path.

Figure 3 runs all three for 3 epochs on the 100-point example, with the same learning rate of 0.05. The race is fair: in every frame, each method reads the same 10 observations. Watch the green mini-batch path: one step per frame, straighter than the orange stochastic path, and far ahead of the red batch path.



**Figure 3:** Batch, mini-batch and stochastic gradient descent reading the same data, 10 observations per frame, for 3 epochs. Left: the paths on the loss contours. Right: the loss against the observations read.

- **Batch** (3 updates in total) has only just started.
- **Stochastic** (300 updates) reaches the minimum, then jumps around it.
- **Mini-batch of 10** (30 updates) follows a much smoother path and is almost there. In its last epoch its path is about as long as the distance it covers (ratio 1.05), while the stochastic path is 27 times longer than its net move.

Each observation's derivative points in a slightly different direction, because each observation carries its own random error. Averaging 10 of them lets these random parts partly cancel, while still updating 10 times as often as batch. The noise of an average of  $B$  values is  $1/\sqrt{B}$  of the noise of one value (Goodfellow §8.1.3), so with  $B = 10$  each step carries about a third (0.32) of a single observation's noise.

## 5. Choosing the batch size

**Key point:** Smaller batches mean more updates per epoch but noisier ones; larger batches mean smoother but fewer updates. A small batch in between, here 8, gets both.

Think of asking for directions: asking one passer-by is quick but may mislead you, polling the whole town is reliable but takes all day, and asking a handful of people is quick and mostly right. Figure 4 trains on the diabetes data with learning rate 0.1 for 100 epochs and four batch sizes, on one train/test split. The score is the **R<sup>2</sup> score** (G-1717) on the test set; ordinary least squares, **OLS** (G-1406), reaches 0.44.



**Figure 4:** Test  $R^2$  per epoch for four batch sizes

The table averages 20 random splits; the last column is how much test  $R^2$  jumps from epoch to epoch near the end (standard deviation over the last 20 epochs).

| Batch size     | Updates per epoch | Test $R^2$ after 100 epochs | Epoch-to-epoch jumps |
|----------------|-------------------|-----------------------------|----------------------|
| 1 (stochastic) | 353               | 0.43                        | 0.076                |
| 8              | 45                | 0.45                        | 0.031                |
| 32             | 12                | 0.40                        | 0.031                |
| 353 (batch)    | 1                 | 0.11                        | 0.006                |

- With batch size 1, test  $R^2$  climbs fastest at first but then jumps from epoch to epoch, more than twice as much as with 8.
- With batch size 8, test  $R^2$  gets close to OLS (0.44) early and stays there more steadily.
- With batch size 32, the curve is smooth but slower, with fewer updates per epoch.
- Full batch has made only 100 updates in total and is far from done.

Like the learning rate, the batch size is tuned by trying values, and the two settings interact. A very small batch gives a noisy derivative, so it may need a small learning rate to stay stable (Goodfellow §8.1.3).

**Extra:** In deep learning, too, small batches often do well: one study got its best results with batch sizes between 2 and 32 (Masters and Luschi). Going the other way, one well-known large-batch method grows the learning rate with the batch size: twice the batch, twice the rate (Goyal et al.).

**Extra:** Batch sizes are often powers of two (16, 32, 64, 128) because some hardware, especially GPUs, runs faster with arrays of these sizes. The memory needed grows with the batch size, not with the size of the dataset, so models can be trained on datasets far

larger than memory (Goodfellow §8.1.3).

## 6. Mini-batch in scikit-learn

**Key point:** SGDRegressor has no batch size option, but its `partial_fit` method trains on one batch at a time, so a short loop gives mini-batch training.

**SGDRegressor** (G-1783, from the previous Note) always updates one observation at a time inside `fit`. Its `partial_fit` method (G-1458), however, does one pass over whatever observations it is given, keeping the coefficients learned so far. Feeding it one batch at a time gives mini-batch-style training:

**Python:** Mini-batch training with `partial_fit`.

```
from sklearn.linear_model import SGDRegressor

sgd = SGDRegressor(learning_rate="constant", eta0=0.2,
                    random_state=0)
rng = np.random.default_rng(0)
for epoch in range(100):
    order = rng.permutation(n)
    for start in range(0, n, 35):
        j = order[start:start + 35]
        sgd.partial_fit(X_train[j], y_train[j])
sgd.score(X_test, y_test)      #  $R^2$  0.45
```

`partial_fit` is also how scikit-learn models learn from data that arrives over time, the **online learning** (G-1391) of Note 5.

**Extra:** Inside `partial_fit`, **SGDRegressor** still updates once per observation of the batch rather than once per batch. True mini-batch updates (one update from the batch’s average derivative) are standard in deep-learning libraries, where the batch size is a basic setting. Those libraries, and many texts, still call the method “SGD” even when each step uses a mini-batch, so “SGD” often means mini-batch gradient descent.

## 7. Summary

|                         | Batch         | Mini-batch                   | Stochastic                   |
|-------------------------|---------------|------------------------------|------------------------------|
| Observations per update | $n$           | a small batch                | 1                            |
| Updates per epoch       | 1             | $n$ / batch size             | $n$                          |
| Path                    | smooth, slow  | fairly smooth, fast          | noisy, fast                  |
| Memory                  | whole dataset | one batch                    | one observation              |
| Typical use             | small data    | most practice, deep learning | very large or streaming data |

- Mini-batch gradient descent includes the other two as batch sizes  $n$  and 1.
- Each epoch: shuffle, cut into batches, update once per batch with the batch's average derivative.
- Batch size and learning rate are tuned together; batch size 8 worked best here (test  $R^2$  0.45, against 0.43 for 1 and 0.40 for 32, averaged over 20 splits).
- In scikit-learn, `partial_fit` on successive batches gives mini-batch-style training.

## 8. Sources

### Built from

- CampusX, “Mini-Batch Gradient Descent”, YouTube, [https://www.youtube.com/watch?v=\\_scscQ4HV](https://www.youtube.com/watch?v=_scscQ4HV)
- StatQuest with Josh Starmer, “Stochastic Gradient Descent, Clearly Explained!!!”, YouTube, <https://www.youtube.com/watch?v=vMh0zPT0tLI>

### Other references

- **Goodfellow**: I. Goodfellow, Y. Bengio, A. Courville, *Deep Learning*, MIT Press, 2016 (deeplearningbook.org). §8.1.3 (batch and minibatch algorithms).
- **Masters and Luschi**: D. Masters and C. Luschi, “Revisiting Small Batch Training for Deep Neural Networks”, arXiv:1804.07612, 2018.
- **Goyal et al.**: P. Goyal et al., “Accurate, Large Minibatch SGD: Training ImageNet in 1 Hour”, arXiv:1706.02677, 2017.

## 9. Key terms

| Term                        | Meaning                                                              |
|-----------------------------|----------------------------------------------------------------------|
| Feature                     | An input variable: one column of the data table                      |
| Observation                 | One record: one row of the data table                                |
| Target                      | The output we predict                                                |
| Epoch                       | One full pass over the training data                                 |
| Mini-batch gradient descent | Gradient descent that updates after each small group of observations |

| Term               | Meaning                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------|
| Batch (mini-batch) | A small group of training observations used for one update                                         |
| Batch size         | The number of observations in each batch; a hyperparameter                                         |
| Shuffling          | Putting the observations in a new random order before each epoch                                   |
| partial_fit        | A scikit-learn method that continues training on new observations, keeping what was learned so far |